



RESTWISE

A Radiation-Free Smartwatch for Student Sleep Wellbeing

PROJECT REPORT

NATIONAL CHILDREN'S SCIENCE CONGRESS
2026

THEME
Food, Agriculture and Health

SUB-THEME
Student Health & Wellbeing

PROJECT TITLE
Restwise — No-Radio Smartwatch for Sleep-Safe Scheduling

SCHOOL / INSTITUTION
[SCHOOL / INSTITUTION NAME]

DISTRICT / STATE
[DISTRICT], [STATE]

TEAM MEMBERS
[TEAM MEMBER 1], [TEAM MEMBER 2], [TEAM MEMBER 3]

GUIDE TEACHER / MENTOR
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REGISTRATION NO.
[REGISTRATION NUMBER]

CERTIFICATE

This is to certify that the project titled “**Restwise — A Radiation-Free Smartwatch for Student Sleep Wellbeing**” has been prepared by

[TEAM MEMBER NAMES]

of [SCHOOL / INSTITUTION NAME], [DISTRICT], [STATE], as an entry for the National Children's Science Congress 2026, under the theme **Food, Agriculture and Health**.

The work embodies the original effort of the above-named participants, carried out under the guidance of the undersigned, and is found satisfactory for submission.

[Guide / Mentor Signature]

[Head of Institution Signature]

*[District Coordinator
Signature]*

Date: [DD/MM/YYYY] Place: [PLACE]

Acknowledgement

We would like to express our sincere gratitude to our guide, **[GUIDE / MENTOR NAME]**, for constant encouragement, technical direction, and patient review through every stage of building Restwise — from the very first survey question to the final working prototype.

We thank the Principal and management of **[SCHOOL / INSTITUTION NAME]** for providing the laboratory access, workspace, and encouragement needed to pursue this project outside the regular curriculum.

We are grateful to the **National Children's Science Congress** and its coordinators at the **[DISTRICT / STATE]** level for the platform and mentorship this competition provides to young innovators.

Our heartfelt thanks go out to the **107 students** whose responses we were able to use, and every other student who took the time to fill out our physical survey honestly and in detail — this project exists because they trusted us with their real daily routines, sleep habits, and stress. We also thank our parents for their support and patience through many late evenings of soldering, coding, and debugging.

Finally, we thank every teacher and friend who piloted an early version of the survey form or gave feedback on the prototype — their honesty made the final product meaningfully better.

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5. Abstract

Rising academic pressure from school, tuition, and homework is widely believed to cut into students' sleep and increase stress, but this effect is rarely measured directly rather than assumed. This study aimed to quantify how students' daily routines affect their sleep duration and stress levels, to identify the specific habits driving the problem, and to design a practical intervention that protects sleep without adding another screen-based device to a child's crowded day.

A physical and digital survey was administered to students across multiple age groups, yielding 107 usable responses from 112 forms submitted, covering their daily schedule and time-block breakdown, self-reported sleep duration on school nights, the frequency and specific triggers of exam-related stress, and their existing break or wind-down habits, if any. Responses were tabulated and analyzed to identify patterns common across different schedules, class levels, and coaching loads, rather than treating each response as an isolated case.

The survey found that 7 in 10 students sleep 8 hours or less on a school night, and a more precise figure computed from reported bed and wake times shows 41.1% sleeping under the 7-hour teen minimum. Nearly 1 in 3 students report feeling stressed by school, tuition, or exams almost every day, and a similar share report never getting a real break between school, tuition, and homework. Combined low and very low daytime energy affects a clear majority (55.1%) of respondents. These findings confirm that sleep shortfall, inconsistent recovery time, and low daytime energy are common, measurable problems within the surveyed student population.

Based on these outcomes, we developed **Restwise**, a low-cost, radiation-free wearable built on an ESP32 microcontroller that deliberately excludes Wi-Fi and Bluetooth, paired with a companion web application. Through a guided process, parents build a sleep-safe schedule that enforces 7–9 hours of sleep and regular breaks during long study periods; this schedule is compiled and transferred to the watch over a wired connection, keeping the child's interaction screen-light.

We conclude that structured, enforced scheduling can meaningfully reduce student sleep deprivation, and recommend wider adoption of low-cost, low-radiation wearables as a practical tool for protecting student health.

6. Introduction

Sleep is one of the most consistently under-protected areas of a student's health. Between school hours, tuition or coaching classes, homework, and daily travel, a modern student's day frequently stretches well past what is considered a healthy schedule for a growing child. Unlike school timings or tuition slots — which are fixed by an institution and rarely negotiable — sleep is the one block of the day with no hard start time, which makes it the block most easily sacrificed when the day runs long.

The National Children's Science Congress 2026 theme, “**Food, Agriculture and Health,**” gave us an opportunity to look closely at student health from an angle that is often overlooked in favour of nutrition and physical fitness: sleep. Good sleep is foundational to a child's memory, mood, immune function, appetite regulation and classroom concentration — arguably one of the least expensive and most effective health interventions available, provided it is actually protected and scheduled.

This project began not from a statistic, but from an everyday observation: students who worked the hardest — attending school, going straight to tuition, and returning home to homework — often appeared the most visibly exhausted. Before attempting to build any kind of solution, we wanted to establish whether this was a genuine, widespread pattern or simply an impression shaped by a few visible cases. This led us to conduct a large-scale physical and digital survey of students, gathering 107 usable responses, the results of which are discussed in detail in later sections of this report.

Guided by what the survey revealed, we designed and built **Restwise**: a low-cost, wearable smartwatch that deliberately avoids Wi-Fi and Bluetooth radios, paired with a companion website that helps a parent build a sleep-safe weekly schedule for their child. This report documents the problem, our research methodology, the survey findings, the technical design of the hardware and software, and the results and future scope of the project.

7. Problem Statement

Students juggling school, tuition/coaching classes, and homework routinely have their sleep compressed to make room for academic demands, with little to no structured recovery time built into their day. This chronic sleep deficit and lack of breaks negatively affects concentration, mood, immunity and long-term health, yet there is no simple, low-cost, distraction-free tool that actively protects a child's sleep and enforces recovery breaks within an already packed schedule.

Existing smartwatches and fitness trackers aimed at this age group typically add another notification-driven, Wi-Fi/Bluetooth-connected screen to the child's day — a device that competes for attention rather than protecting rest. There is a clear need for a device that enforces a healthy schedule *without* becoming a new source of distraction itself.

8. Objectives

The objectives of this project are:

- 1 To measure, through a physical and digital survey of students (107 usable responses), how daily academic routines affect sleep duration, stress frequency, and break/wind-down habits.
- 2 To design and build a low-cost wearable device that enforces a minimum healthy sleep duration (7–9 hours) and inserts regular breaks into long study stretches.
- 3 To ensure the device carries no wireless radios (no Wi-Fi, no Bluetooth) so that it cannot itself become a source of notifications or distraction for the child.
- 4 To build a simple, guided companion web application that lets a parent plan, review and update the child's weekly schedule without any technical knowledge.
- 5 To validate the sleep-safety logic (minimum sleep hours, break insertion) in software before any schedule reaches the watch.
- 6 To evaluate the feasibility of the prototype for a wider, low-cost rollout to more families and schools.

9. Hypothesis / Research Question

Research Question: Does a structured, enforced daily schedule — delivered through a low-cost, radiation-free wearable — measurably help protect a student's minimum required sleep duration and encourage regular breaks, compared to an unmanaged daily routine?

Hypothesis (H1): Students whose daily routines are structured around an enforced sleep-safe schedule (7–9 hours of sleep, with regular breaks between study blocks) will report improved sleep adequacy and reduced perceived stress compared to their current, unmanaged routines.

Null Hypothesis (H0): There is no measurable difference in sleep adequacy or perceived stress between students following an unmanaged routine and students using an enforced sleep-safe schedule.

This hypothesis is intended to be tested formally in the trial phase described in **Section 24 – Future Scope**; the survey documented in this report establishes the baseline problem the hypothesis responds to.

10. Survey Methodology

To avoid designing a solution around assumption, we conducted a structured, anonymous-optional **physical (paper) survey** before any hardware or software design began.

Sample size

112 forms were submitted, spanning multiple classes/grades and age groups, across [SCHOOL(S) / LOCATION(S)]; **107 responses** were complete and usable for analysis after removing incomplete or invalid submissions.

Survey type

A printed, single-page form, filled out by hand. Respondents could choose to answer anonymously, in which case name and school fields were left blank by design.

Data captured

- Respondent's daily schedule and time-block breakdown (school, tuition, homework, travel).
- Self-reported sleep duration on a typical school night.
- Frequency and specific triggers of exam or school-related stress.
- Existing break or wind-down habits, if any, between study blocks.

Data handling

Responses were manually compiled into a structured digital format for tabulation and analysis. Anonymous responses were kept anonymous throughout — no attempt was made to identify a respondent from their handwriting or answers.

11. Survey Questionnaire

The questionnaire consisted of eight questions — four multiple-choice and four short-answer — designed to be completed in under two minutes, along with fields for Name (optional / skippable if anonymous), Class, and School.

No.	Question	Response format
Q1	How many hours of sleep do you usually get on a school night?	(a) Less than 5 hours (b) 6–8 hours (c) More than 8 hours
Q2	How often do you feel stressed because of school, tuition, or exams?	(a) Almost every day (b) Sometimes (c) Never
Q3	Do you get a proper break between school, tuition, and homework?	(a) Always (b) Sometimes (c) Never
Q4	How would you describe your energy level during the day?	(a) Very low (b) Low (c) Average (d) High
Q5	What time do you usually go to sleep?	Short answer
Q6	What time do you usually wake up?	Short answer
Q7	How many days a week do you go to tuition / coaching?	Short answer
Q8	Which tuition / coaching classes do you attend?	Short answer

12. Survey Results – Tables

The tabulated results across the 107 usable responses (from 112 forms submitted) are summarised below.

Table 12.1 — Sleep duration (self-reported, Q1)

Sleep duration on a school night (self-reported)	% of respondents	Count
Less than 5 hours	30%	32
6 – 8 hours	39%	42
More than 8 hours	31%	33

Table 12.1b — Sleep duration computed from reported bed/wake times (Q5/Q6) — average 7.18 hrs; 41.1% of respondents fall under the 7-hour teen minimum

Sleep duration (computed, Q5/Q6)	% of respondents	Count
Less than 5 hours	3.7%	4
5 – 6 hours	13.1%	14
6 – 8 hours	59.8%	64
More than 8 hours	23.4%	25

Table 12.2 — Stress frequency

Stress frequency (school / tuition / exams)	% of respondents	Count
Almost every day	31.8%	34
Sometimes	28.0%	30
Never	40.2%	43

Table 12.3 — Break habits

Breaks between school, tuition & homework	% of respondents	Count
Never	30.8%	33
Sometimes	37.4%	40
Always	31.8%	34

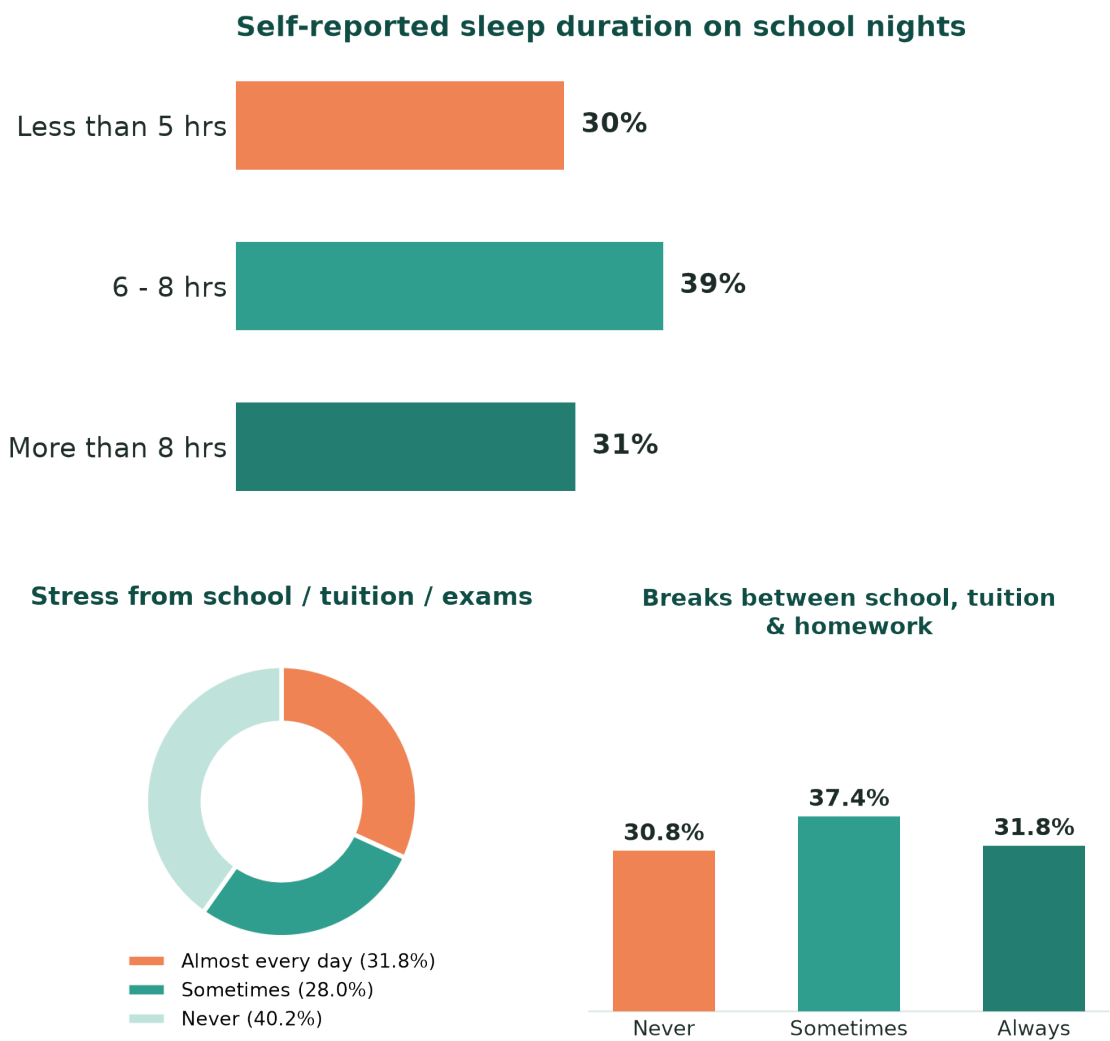
Table 12.4 — Daytime energy level

Energy level during the day	% of respondents	Count
Very low	29.9%	32
Low	25.2%	27
Average	22.4%	24
High	22.4%	24

Source: 107 usable digital survey responses (of 112 forms submitted), tabulated via the Restwise admin export. Percentages are rounded to one decimal place; totals may not sum to exactly 100% for this reason.

13. Survey Results – Graphs/Charts

The same data is presented visually below for quicker interpretation.



14. Analysis of Survey Results

Across the 107 usable responses, the data points to a real but more nuanced picture than a single “everyone is exhausted” headline — sleep shortfall is common though not universal, and stress and recovery time are split roughly into thirds rather than concentrated in a large majority.

1. A meaningful share of students fall short on sleep

Self-reported answers show 7 in 10 students (69%) sleep 8 hours or less on a school night, and nearly 3 in 10 (30%) sometimes sleep under 5 hours. The more precise figure, computed directly from each student's reported bed and wake time rather than a self-chosen category, puts the average at 7.18 hours and finds 41.1% below the 7-hour teen minimum — smaller than the self-reported figure suggests, but still affecting more than 2 in 5 students.

2. Stress is common but not the majority experience

31.8% of respondents — nearly 1 in 3 — report feeling stressed by school, tuition, or exams almost every day, while 40.2% report never feeling this way. Stress from academic workload is therefore a significant, but not dominant, experience across the surveyed group.

3. Recovery breaks are inconsistent, not absent

30.8% of respondents — again nearly 1 in 3 — report never getting a real break between school, tuition, and homework, while a plurality (37.4%) get one only sometimes. Only 31.8% consistently get a proper break, meaning most students' recovery time depends on an inconsistent day-to-day routine rather than a reliable structure.

4. Low daytime energy is the majority experience

Very low and low energy combined account for 55.1% of respondents — more than half — making daytime energy the single measure where a clear majority of students report a negative outcome, even though sleep shortfall and stress individually affect smaller shares.

Implication for design

These findings directly shaped Restwise's two core enforcement rules: a minimum sleep duration check (7–9 hours, with only the day's longest sleep block required to meet the minimum), and automatic insertion of short water/rest breaks during long study stretches — targeted specifically at the roughly 1-in-3 students getting no consistent break and the 2-in-5 falling short of the 7-hour minimum.

15. Materials Used – Hardware

A deliberate design constraint of this project was to use only low-cost, easily sourced components — the goal was a buildable prototype, not a theoretical concept.

Component	Purpose	Approx. Qty
ESP32 development board	Main microcontroller — Wi-Fi/Bluetooth capable but deliberately left switched off and unused	1
Small display (OLED / LCD)	Shows current schedule block and time to the student	1
Buzzer / vibration motor	Alerts for breaks, bedtime, and schedule transitions	1
Push button(s)	Basic on-device interaction (acknowledge alert, view next block)	2–3
Rechargeable Li-Po battery + charging module	Portable, wearable power source	1
USB Micro/Type-C cable	The <i>only</i> data link — used to transfer the compiled schedule from the companion website to the watch	1
3D-printed / laser-cut enclosure	Wearable case and strap mounting	1 set
Connecting wires, resistors, prototyping board	Circuit assembly	as needed

*Note on radios: the ESP32 is chosen specifically for its low cost, wide availability, and ease of development — **not** for its Wi-Fi/Bluetooth capability. Both radios are left disabled in firmware and never activated at any point in normal operation.*

16. Materials Used – Software

The software stack was chosen to keep the whole system free of any subscription cost, and to keep the watch itself entirely offline after each weekly update.

Tool / Technology	Role in the project
Arduino IDE / ESP-IDF (C/C++)	Firmware for the ESP32 — reads the compiled schedule, drives the display and buzzer, tracks time offline
Python 3 + Flask	Backend web server for the companion website and its APIs
Supabase (PostgreSQL, Auth, Storage)	Cloud database, user accounts, and file storage — access controlled entirely through Row Level Security policies
Groq API (LLM)	Powers the guided, conversational onboarding chat that builds a parent's schedule
HTML5 / CSS3 / vanilla JavaScript	Front-end of the companion website (landing page, dashboard, timetable editor, admin console)
Web Serial API	Browser-native, driver-free USB communication used to upload the compiled schedule to the watch
PyMuPDF (fitz)	Fills and exports the printed survey response as a PDF certificate
RWBIN (custom binary protocol)	A compact binary schedule format designed for this project, so a full week's timetable fits in a few hundred bytes
Vercel	Hosting platform for the companion website

17. Working Principle

Restwise operates on a simple, weekly loop with three stages:

1. Plan

A parent visits the companion website and has a short, guided conversation (powered by an LLM) that asks about the child's real school timings, tuition/coaching schedule, homework routine, and bedtime — one question at a time, in plain language.

2. Check

Before the schedule can be saved, the system automatically checks it against two rules: (a) the day's longest sleep block must be at least 7 hours (target range 7–9 hours), and (b) any continuous study stretch beyond a set interval automatically gets a short water/rest break inserted. A schedule that fails either check is flagged back to the parent before it can be sent to the watch.

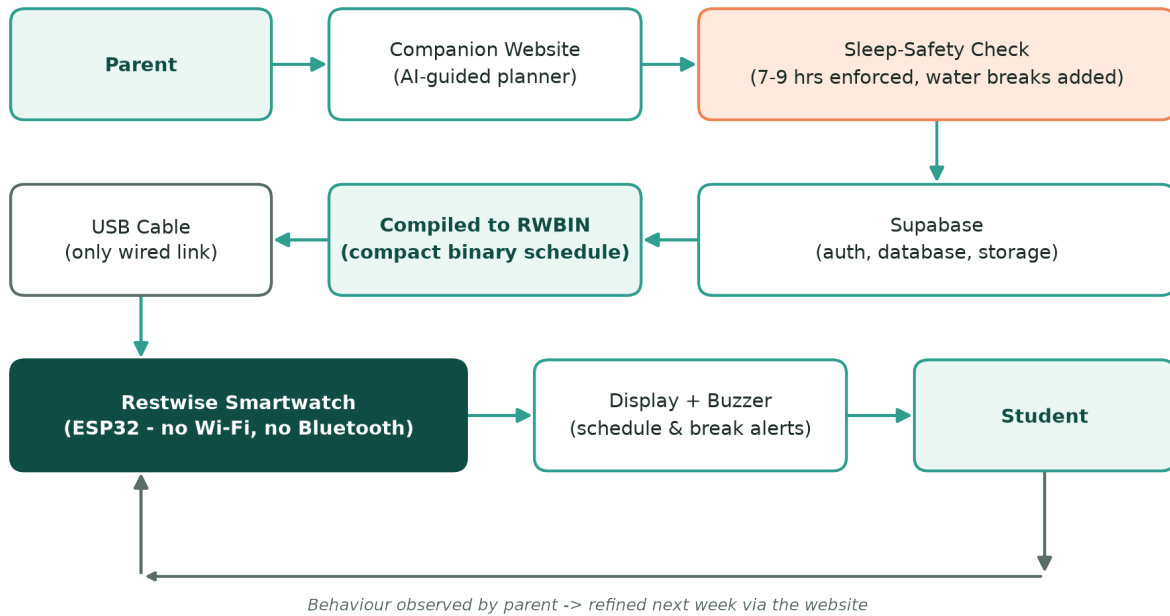
3. Sync

Once approved, the schedule is compiled into RWBIN, a compact custom binary format, and transferred to the watch over a single USB cable using the browser's Web Serial API — no companion app, no pairing, and no wireless connection of any kind. After the cable is disconnected, the watch runs entirely offline for the week, using its own onboard clock to track the schedule and trigger display updates and buzzer alerts at the right times.

This weekly plan → check → sync loop means the intelligence and connectivity live entirely on the website side, while the watch itself stays deliberately “dumb,” offline, and free of any radio hardware.

18. Block Diagram / Flowchart

The diagram below traces one full weekly cycle, from the parent's input through to the student wearing the watch, and the feedback loop back into the following week's plan.



19. Smartwatch Design / Prototype

The Restwise prototype is built around an ESP32 development board housed in a compact, wrist-mountable enclosure. Its design deliberately favours simplicity over feature density:

- A small display shows only what matters right now — the current schedule block and the time remaining in it.
- A single buzzer/vibration motor provides a short, distinct alert for breaks, meals, and bedtime — no notification feed, no messages.
- One or two physical buttons allow the student to acknowledge an alert or glance at the next block, with no touchscreen and no app to open.
- The only external port is a USB connector, used exclusively for charging and for the weekly schedule transfer — never for any other data exchange.
- The Wi-Fi and Bluetooth radios present on the ESP32 chip are left switched off in firmware at all times; this is a firmware-level and physical design decision, not a configuration toggle the user can accidentally enable.

[Insert prototype photographs here — front view, side view, and the watch connected via USB during a schedule upload.]

20. Website Design & Features

The companion website is the “brain” of the system — all planning, checking, and account management happens here, keeping the watch itself simple and offline.

Landing page

Explains the problem and the product, and hosts the public physical-survey digital submission form used to collect the data discussed in Sections 10–14 of this report.

Accounts

Secure sign-up, log-in, and password reset for parents, backed by Supabase Auth.

Guided AI onboarding

A conversational, one-question-at-a-time chat that builds a first schedule for a new family, explained in Section 17.

Dashboard (View / Edit Timetable)

A always-available weekly timetable view, and a manual editor for updating individual blocks after the first schedule is built — both re-run the same sleep-safety and break checks before saving.

Manage Smartwatch

Compiles the current timetable into the RWBIN binary format and uploads it to the watch over USB, with version tracking so a parent can see when the watch was last updated.

Admin Console

A role-gated area for reviewing and exporting survey submissions (with an optional data-masking mode that replaces identifying details with placeholders for privacy), used to compile the results in this report.

21. Testing and Observations

Testing was carried out at three levels: the sleep-safety logic in isolation, the end-to-end website flow, and the physical watch-website communication.

Logic testing

The sleep-adequacy and break-insertion rules were tested against a range of sample schedules — including edge cases such as split sleep across a nap and a main sleep block — to confirm that only the day's longest continuous sleep block is required to meet the 7-hour minimum, as intended.

End-to-end website testing

The full account → onboarding → dashboard → compile → upload flow was tested repeatedly during development, including a seeded batch of survey responses used to verify the admin console's statistics, export, and data-masking features under realistic data volumes.

Hardware / communication testing

The USB upload path (browser → Web Serial API → ESP32) was tested for both first-time device pairing and subsequent, already-authorized reconnections, confirming that a schedule update can be sent reliably without requiring a native driver installation.

Key observation

Deployment testing on cloud hosting surfaced a real platform constraint — serverless functions impose a hard cap on request/response size — which shaped a design decision to route large file transfers (survey scans, PDF exports) directly between the browser and cloud storage rather than through the web server, avoiding that limit entirely.

22. Results

This project produced two categories of results: findings from the survey, and outcomes from building the prototype itself.

Survey findings

- 7 in 10 (69%) surveyed students sleep 8 hours or less on a school night; a more precise figure computed from reported bed/wake times shows 41.1% sleeping under the 7-hour teen minimum.
- Nearly 1 in 3 (31.8%) report feeling stressed almost every day due to school, tuition, or exams.
- Nearly 1 in 3 (30.8%) report never getting a real break between school, tuition, and homework.
- More than half (55.1%) report low or very low daytime energy — the single measure where a clear majority of students report a negative outcome.

Prototype outcomes

- A working ESP32-based smartwatch prototype that displays the current schedule block and alerts for breaks and bedtime, with no Wi-Fi or Bluetooth radio active at any point.
- A functioning companion website covering sign-up/login, AI-guided onboarding, a timetable dashboard, smartwatch sync, and an admin console for survey management.
- A working sleep-safety and break-insertion check that runs automatically on every schedule before it can be sent to the watch.
- A custom, compact binary protocol (RWBIN) that fits a full week's schedule into a small enough payload for fast, reliable USB transfer.

Together, these results demonstrate that the core hypothesis of this project — that a low-cost, radio-free wearable can enforce a sleep-safe schedule without adding another screen-based distraction — is technically achievable with entirely off-the-shelf components.

23. Advantages & Limitations

Advantages	Limitations
Zero radio emission — no Wi-Fi/Bluetooth, so the watch cannot notify, track, or distract the child.	Requires a wired USB connection to a computer once a week to sync a new schedule.
Built entirely from low-cost, widely available components — no proprietary or expensive parts.	Currently a single hand-built prototype; not yet tested for durability or comfort across a full school term.
No subscription or ongoing internet connection required on the child's side.	The schedule quality depends on the accuracy of what the parent enters into the onboarding chat.
The sleep-safety and break rules are enforced in software, removing reliance on willpower alone.	The device has not yet been trialled with real families over an extended period to measure actual sleep improvement.
Privacy-conscious data handling — row-level access control rather than open data storage.	The current prototype supports one fixed alert pattern (buzzer); richer feedback is future work.

24. Future Scope & Conclusion

Future scope

- 1 **Trial with real families** — deploy the prototype to a small group of families for a month and measure whether reported sleep duration and stress actually improve.
- 2 **Smarter, gentler wake-up** — use simple movement sensing to wake the child at a lighter point in their sleep cycle rather than at a fixed alarm time.
- 3 **School edition** — a class-level dashboard so a school counsellor or teacher can observe aggregate sleep/stress patterns and intervene early.
- 4 **Wider hardware validation** — test comfort, battery life, and durability of the enclosure across a full school term with multiple prototype units.
- 5 **Larger-scale survey** — extend the study beyond [SCHOOL / DISTRICT] to build a more statistically robust picture of the problem across different regions and school types.

Conclusion

This project set out to test a simple premise: that a large share of students are sleep-deprived not through a lack of awareness, but because nothing in their day actively protects their sleep. Our survey of 107 students confirmed this at meaningful scale, and Restwise — a low-cost, radiation-free smartwatch paired with a guided companion website — demonstrates that enforcing a sleep-safe schedule is technically achievable without introducing another screen-based distraction into a child's life. Protecting sleep costs nothing beyond attention and structure; this project's contribution is a small, buildable tool that supplies exactly that structure.

25. References + Appendix

References

- 1 National Sleep Foundation — Sleep duration recommendations for teenagers and adolescents.
- 2 American Academy of Pediatrics — Policy statement on school start times and adolescent sleep.
- 3 World Health Organization — Guidelines on physical activity, sedentary behaviour and sleep for children and adolescents.
- 4 Espressif Systems — ESP32 Technical Reference Manual.
- 5 MDN Web Docs — Web Serial API documentation.
- 6 Supabase Documentation — Row Level Security and Authentication.
- 7 [Add any additional sources, textbooks, or papers consulted, in your school's preferred citation style.]

Appendix

- Appendix A — Blank copy of the physical survey questionnaire.
- Appendix B — Full tabulated survey response data.
- Appendix C — Firmware and website source code (repository link: **restwise-ncsc.vercel.app**).
- Appendix D — Prototype photographs and circuit diagram.

— End of Report —